

Unit 6, Lesson 13: Similarity in the Real World



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems involving similarity in two-dimensional figures.



6.13.1 Warm-Up: With Math I CAN

1. What struggles have you had in mathematics?
2. Students often ask teachers, “When will I ever use this?” What is one thing in mathematics you’ve wondered if you would ever use?
3. Sometimes it isn’t that a job requires you to prove a theorem, but to use the logical thinking you learned in proving a theorem. Describe a few careers that you’re interested in and what can you do with math.



6.13.2 Activity: Real-World Problem Gallery Walk

Complete each problem by recording your work and answer in the spaces provided.

Problem	Work	Answer
Problem A		
Problem B		

Problem C		
Problem D		



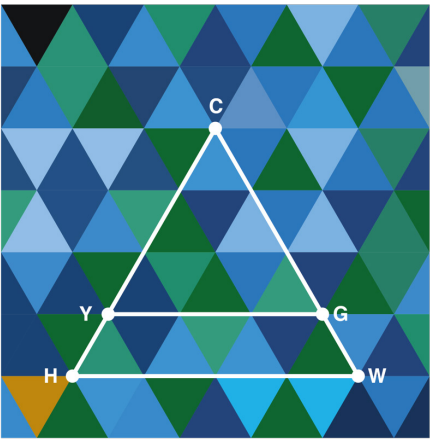
Problem E		
Problem F		



6.13.3 Wrap-Up: Ticket Out the Door

- Beth and Trevor are trying to find the lengths of the triangles on the quilt.

Beth and Trevor know that $\triangle CYG \sim \triangle CHW$, $CG = 21$ inches (in), $YG = (2.5x + 1.5)$ in, $GW = 7$ in, and $HW = (6x - 14)$ in. Their work to find the scale factor is shown.



Beth's Work	Trevor's Work
$\frac{CG}{GW} = \frac{21}{7}$ Scale Factor is 3	$\frac{CG}{CW} = \frac{21}{28}$ Scale Factor is 0.75

- Whose work do you agree with?
- What is the length of $\overline{HW}$?